4月14日. 周一. 第16次课.

Topic1: 定积分的换元法与分部积分法.(续).

(3) 定积分的分部积分法.(Integration by Parts).

[中文课本: 6.2 定积分的分部积分法. P229~P232; Calculus: P475~P476]

$$\int_a^b u(x)\cdot v'(x)\,dx = u(x)\cdot v(x)\Big|_a^b - \int_a^b u'(x)\,v(x)\,dx.$$

Hw!

证明: 直接使用微积分基本定理即可. 留作习题.

e.g.7. $\int_1^2 x\cdot\ln x\cdot dx = \int_1^2 \ln x\cdot \frac{d(x^2)}{2} = \left(\ln x\cdot\frac{x^2}{2}\right)\Big|_1^2 - \int_1^2 \frac{x^2}{2}\cdot d(\ln x)$

$= 2\ln 2 - 0\cdot\frac{1}{2} - \int_1^2 \frac{x}{2}\cdot dx = 2\cdot\ln 2 - \left(\frac{x^2}{4}\right)\Big|_1^2$

$= 2\cdot\ln 2 - \left(\frac{4}{4}-\frac{1}{4}\right) = 2\cdot\ln 2 - \frac{3}{4}.$

e.g.8. $\int_0^1 \arctan x\cdot dx = \left(x\cdot\arctan x\right)\Big|_0^1 - \int_0^1 x\cdot d(\arctan x)$

$= 1\cdot\frac{\pi}{4} - 0\cdot 0 - \int_0^1 \frac{x}{1+x^2}\,dx = \frac{\pi}{4} - \int_0^1 \frac{\frac{1}{2}\cdot d(1+x^2)}{1+x^2}.$

$\xlongequal{t=1+x^2} \frac{\pi}{4} - \int_1^2 \frac{dt}{2t} = \frac{\pi}{4} - \left(\frac{1}{2}\ln|t|\right)\Big|_1^2 = \frac{\pi}{4} - \frac{1}{2}\ln 2.$

讲不定积分时提到的"循环现象"与"递推现象仍会出现.

e.g.9. 求 $I = \int_0^{\pi/2} e^{3x}\cdot\cos 2x\,dx.$

$I = \int_0^{\pi/2} e^{3x}\cdot\cos 2x\,dx = \int_0^{\pi/2}\cos 2x\cdot\frac{d(e^{3x})}{3}.$

$= \left(\cos 2x\cdot\frac{e^{3x}}{3}\right)\Big|_0^{\pi/2} - \int_0^{\pi/2}\frac{e^{3x}}{3}\cdot d(\cos 2x)$

$= \cos\pi\cdot\frac{e^{3\pi/2}}{3} - \cos 0\cdot\frac{e^0}{3} - \int_0^{\pi/2}\frac{e^{3x}}{3}\cdot(-\sin 2x\cdot 2)\,dx$

$= -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3} + \int_0^{\pi/2}\frac{2}{3}\cdot e^{3x}\cdot\sin 2x\,dx$

$= -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3} + \int_0^{\pi/2}\frac{2}{3}\cdot\sin 2x\cdot\frac{d(e^{3x})}{3}$

$= -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3} + \left(\frac{2}{3}\cdot\sin 2x\cdot\frac{e^{3x}}{3}\right)\Big|_0^{\pi/2} - \int_0^{\pi/2}\frac{e^{3x}}{3}\cdot d\left(\frac{2}{3}\cdot\sin 2x\right)$

$= -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3} + 0 - 0 - \int_0^{\pi/2}\frac{e^{3x}}{3}\cdot\frac{2}{3}\cdot\cos 2x\cdot 2\,dx$

$= -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3} - \frac{4}{9}I$

$\therefore \frac{13}{9}I = -\frac{1}{3}\cdot e^{3\pi/2} - \frac{1}{3}$ $\therefore I = -\frac{3}{13}e^{3\pi/2} - \frac{3}{13}$

eg.10. 求 $I_n=\int_0^{\pi/2}\sin^n x\,dx.\ (n\geq 0).$

$$=\int_0^{\pi/2}\sin^{n-1}x\cdot(\sin x\,dx)=\int_0^{\pi/2}\sin^{n-1}x\cdot d(-\cos x).$$

$$=\sin^{n-1}x\cdot(-\cos x)\Big|_0^{\pi/2}-\int_0^{\pi/2}(-\cos x)\cdot d\cdot(\sin^{n-1}x).$$

$$\sin^{n-1}x\cdot(-\cos x)\Big|_0^{\pi/2}=0-\sin^{n-1}0\cdot(-\cos 0)\begin{cases}1. & n=1\text{时}\\ 0. & n\geq 2\text{时}\end{cases}$$

$$-\int_0^{\pi/2}(-\cos x)\cdot d(\sin^{n-1}x)=\int_0^{\pi/2}\cos x\cdot(n-1)\cdot\sin^{n-2}x\cdot\cos x\cdot dx$$

$$=(n-1)\cdot\int_0^{\pi/2}\sin^{n-2}x\cdot\cos^2 x\,dx$$

$$=(n-1)\cdot\int_0^{\pi/2}\sin^{n-2}x\cdot(1-\sin^2x)\,dx$$

$$=(n-1)\cdot I_{n-2}-(n-1)\cdot I_n$$

如果 $n\geq 2$ 时. 有 $I_n=0+(n-1)\cdot I_{n-2}-(n-1)\cdot I_n$

$$\therefore\ n\cdot I_n=(n-1)\cdot I_{n-2}.\qquad \therefore I_n=\frac{n-1}{n}\cdot I_{n-2}$$

$I_0=\int_0^{\pi/2}\sin^0x\,dx=\pi/2.\qquad I_1=\int_0^{\pi/2}\sin x\,dx=(-\cos x)\Big|_0^{\pi/2}=1$

$I_2=\frac{1}{2}\cdot I_0=\pi/4.\qquad I_3=\frac{2}{3}I_1=\frac{2}{3}.$

$I_4=\frac{3}{4}\cdot I_2=\frac{3}{16}\pi.\qquad I_5=\frac{4}{5}I_3=\frac{8}{15}$

$I_2=\frac{1}{2}\cdot I_0,\quad I_4=\frac{3}{4}I_2=\frac{3}{4}\cdot\frac{1}{2}\cdot I_0.\quad I_6=\frac{5}{6}I_4=\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot I_0$

一般地. $I_{2n}=\frac{2n-1}{2n}\cdot\frac{2n-3}{2n-2}\cdots\frac{1}{2}\cdot I_0=\left(\frac{(2n-1)!!}{(2n)!!}\cdot\frac{\pi}{2}\right)$

$I_3=\frac{2}{3}\cdot I_1.\quad I_5=\frac{4}{5}I_3=\frac{4}{5}\cdot\frac{2}{3}\cdot I_1\quad I_7=\frac{6}{7}\cdot I_5=\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot I_1$

$$I_{2n-1}=\frac{2n-2}{2n-1}\cdot\frac{2n-4}{2n-3}\cdots\frac{2}{3}\cdot\frac{I_1}{1}\quad\left(=\frac{(2n-2)!!}{(2n-1)!!}\right)$$

双阶乘:(!!). n!! 表示 ≤n 的.且与n奇偶性相同的数的乘积

e.g. $6!!=6\times4\times2=48.\quad 7!!=7\times5\times3\times1=105.$

e.g. 10+. 求 $J_n = \int_0^{\pi/2} \cos^n x \, dx$.

奇变偶不变，符号看象限.

$\xlongequal{t=\frac{\pi}{2}-x} \int_{\pi/2}^{0} \cos^n\left(\frac{\pi}{2}-t\right) \cdot d\left(\frac{\pi}{2}-t\right)$

$\cos\left(\frac{\pi}{2}-t\right) = \sin t$.

$= \int_{\pi/2}^{0} \sin^n t \, (-dt)$

$= \int_0^{\pi/2} \sin^n t \cdot dt = \int_0^{\pi/2} \sin^n x \cdot dx = I_n$

e.g. $J_6 = \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot J_0 = \frac{5\pi}{32}$.

Topic2: 利用对称性计算定积分.

[中文课本: 6.3. 利用对称性计算定积分. $P_{235} \sim P_{241}$

Calculus: Symmetry. $P_{417} \sim P_{418}$.]

在上一个 Topic 中我们看到, 只要能求出 $f(x)$ 的不定积分,
就能求出 $f(x)$ 的定积分.

在这个 Topic 我们会看到, 对某些难以求不定积分的 $f(x)$,
也可以用一些技巧求定积分.

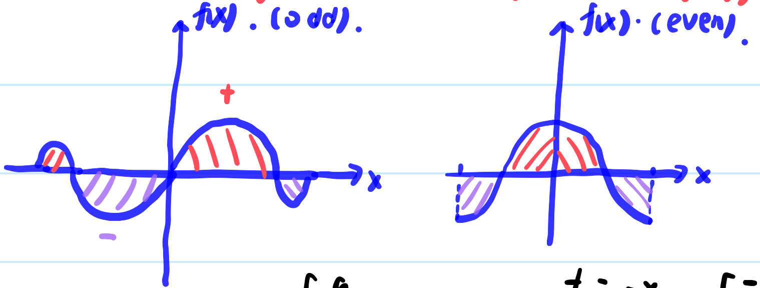
因此我们可以说, 求定积分往往比求不定积分容易.

(1). 奇函数 / 偶函数 的定积分.

定理1. 已知 $f(x)$ 是 $[-a, a]$ 上的可积函数.

<1>. 若 $f(x)$ 是奇函数. ($f(-x) = -f(x)$), 则 $\int_{-a}^{a} f(x)\,dx = 0$.

<2>. 若 $f(x)$ 是偶函数. ($f(-x) = f(x)$), 则 $\int_{-a}^{a} f(x) = 2 \cdot \int_{0}^{a} f(x) \cdot dx$.



证明: <1> $I = \int_{-a}^{a} f(x)\,dx \overset{t=-x}{=\!=\!=} \int_{a}^{-a} \cdot f(-t) \cdot d(-t) = \int_{a}^{-a} (-f(t))(-dt)$

$= \int_{a}^{-a} \cdot f(t) \cdot dt = -\int_{-a}^{a} f(t)\,dt = -I$.

$\because I = -I \quad \therefore I = 0$

题. <2> 留作习题. $\int_{-a}^{a} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$

所以在关于原点对称的区间 $[-a, a]$ 上, 奇, 偶函数的积分可以简化.

e.g.1. $\int_{-2}^{2} (x^6 + 1) \cdot dx$.

$= 2 \cdot \int_{0}^{2} (x^6 + 1) \cdot dx = 2 \cdot \left(\frac{x^7}{7} + x\right)\Big|_{0}^{2} = 2 \cdot \left(\frac{2^7}{7} + 2\right) = \frac{2^8}{7} + 4 = \frac{\sim}{7}$

$f(x)_{11}$

$f(-x)=\frac{\tan(-x)}{1+(-x)^2+(-x)^4}=\frac{-\tan x}{1+x^2+x^4}=-f(x)$ $\therefore f(x)$ is odd

e.g. 2. $\int_{-1}^{1}\frac{\tan x}{1+x^2+x^4}\,dx = 0$ (因为$f(x)$是奇函数).

偶函数. 奇函数.

e.g. 3. $\int_{-\pi/2}^{\pi/2}(\cos^5 x + x\cdot\sin^4 x)\,dx$

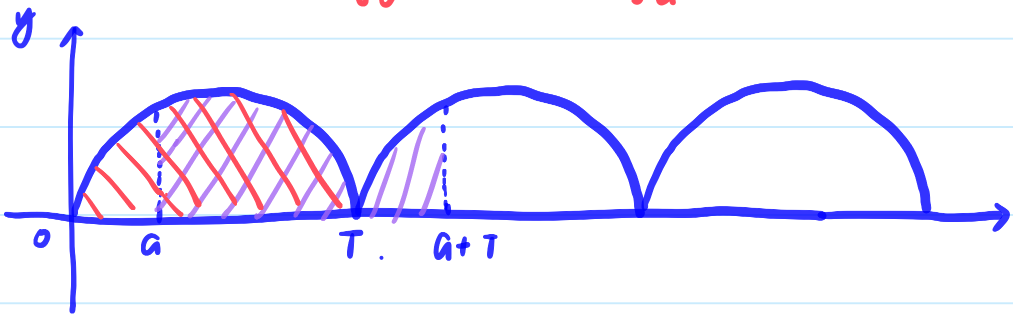
$= 2\int_0^{\pi/2}\cos^5 x\,dx + 0$

$= 2\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot\frac{1}{1} = \frac{16}{15}$

(2). 周期函数的定积分.

定理2: 设$f(x)$是$(-\infty,+\infty)$上的连续函数，且以T为周期.($f(x+T)=f(x)$, $\forall x$).

则 $\forall a\in(-\infty,+\infty)$, 有 $\int_0^T f(x)\,dx = \int_a^{a+T} f(x)\,dx$.



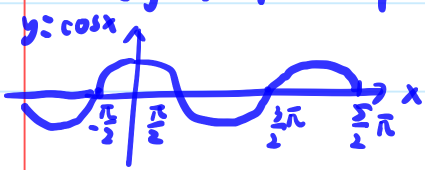
周期为T的函数 在长度为T的区间上积分结果总一样.(和上限,下限无关).

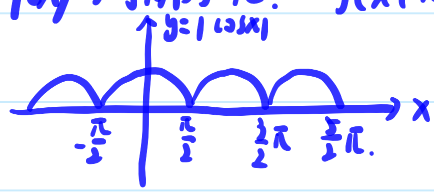
e.g. 4. $\int_1^{1+\pi}|\cos x|\cdot dx$.

记$f(x)=|\cos x|$. 则 $f(x)$ 周期为 π. $f(x+\pi)=|\cos(x+\pi)|=|-\cos x|$

$=|\cos x|$

$=f(x)$. $(\forall x\in R)$





$\therefore \int_1^{1+\pi}|\cos x|\,dx = \int_{-\pi/2}^{\pi/2}|\cos x|\,dx = \int_{-\pi/2}^{\pi/2}\cos x\cdot dx$

$=(\sin x)\Big|_{-\pi/2}^{\pi/2} = 1-(-1) = 2$.

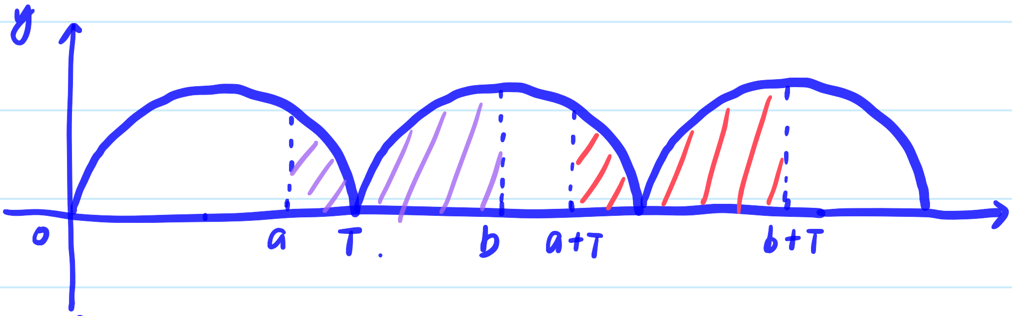
定理2 的证明:

$\int_a^{a+T}f(x)\cdot dx = \int_a^T f(x)\,dx + \int_T^{a+T}f(x)\,dx$ 而 $\int_T^{a+T}f(x)\,dx \overset{t=x-T}{=\!=} \int_0^a f(t+T)\cdot d(t+T) = \int_0^a f(t)\cdot dt$

$= \int_a^T f(x)\,dx + \int_0^a f(x)\,dx = \int_0^T f(x)\,dx$

定理3：设 $f(x)$ 是 $(-\infty,+\infty)$ 上的连续函数，且以T为周期. ($f(x+T)=f(x), \forall x$).

则 $\forall a,b\in(-\infty,+\infty)$，有 $\int_{a+T}^{b+T} f(x)\,dx = \int_a^b f(x)\,dx$.

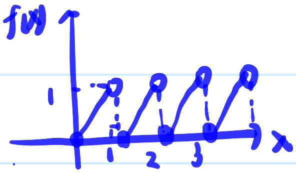


e.g.5. 求 $\int_0^n (x-[x])\,dx$. $(n\in N_+)$

($[x]$ 表示 $\leq x$ 的最大整数. e.g. $[1]=1$, $[1.5]=1$, $[-1.5]=-2$)

$f(x)=x-[x]$ 是一个周期为1的函数.

$f(x+1)=x+1-[x+1]=(x+1)-([x]+1)=x-[x]=f(x)$



$\therefore \int_0^n (x-[x])\,dx = \int_0^1 f(x)\,dx + \int_1^2 f(x)\,dx + \cdots + \int_{n-1}^n f(x)\,dx$

$= \int_0^1 f(x)\cdot dx + \int_0^1 f(x)\,dx + \cdots + \int_0^1 f(x)\,dx$ (共n个)

$= n\cdot\int_0^1 f(x)\,dx = n\cdot\int_0^1 x\cdot dx = \frac{n}{2}$.

定理3的证明：

$$\int_{a+T}^{b+T} f(x)\,dx \xlongequal{t=x-T} \int_a^b f(t+T)\cdot d(t+T) = \int_a^b f(t)\cdot dt = \int_a^b f(x)\cdot dx$$

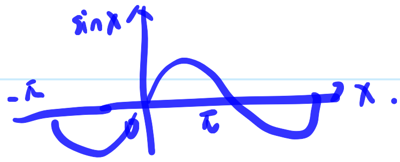
$\boxed{\text{Try it!}}$

$\int_{-\pi}^{\pi} \frac{\sin x}{\sqrt{1+x^4}}\,dx = 0$. (奇函数).

$\int_0^{2025\pi} |\sin x|\,dx = 2025\cdot\int_0^{\pi} \sin x\,dx$

$= 2025\cdot(-\cos x)\Big|_0^{\pi} = 4050$.

$f(x)=|\sin x|$. $f(x+\pi)=|\sin(x+\pi)|=|-\sin x|=|\sin x|=f(x)$



$\int_{2\pi}^{5\pi/2} \sin^5 x\,dx$

$= \int_0^{\pi/2} \sin^5 x\,dx$

$= \frac{4}{5}\cdot\frac{2}{3}\cdot\frac{1}{1} = \frac{8}{15}$